

Used Car Price Prediction

Exploratory Data Analysis & Feature Engineering Report

Name	Manahil
Internship	LearnDepth
Project	Problem 9 — Used Car Price (EDA + Feature Engineering Pipeline)
Dataset	09_used_car_price.csv
Tools Used	Python, pandas, numpy, matplotlib, seaborn (Google Colab)
Target Variable	price_lakh

1. Problem Statement

The goal of this project is to analyze used-car characteristics — such as age, kilometers driven, engine size, and number of owners — to understand the factors associated with a car's resale price (price_lakh). The dataset contains 1000 records with five columns: car_age_years, km_driven, engine_cc, owners, and price_lakh.

2. Tools and Libraries

- pandas — loading and manipulating tabular data
- numpy — numerical operations
- matplotlib — plotting
- seaborn — statistical visualizations (histograms, boxplots, heatmaps)

Environment: Google Colab

3. Dataset Inspection

The dataset was loaded using `pandas.read_csv()` and inspected using `.shape`, `.columns`, `.dtypes`, and `.describe()` to understand its structure, size, and basic statistics before any analysis began.

```
df = pd.read_csv("09_used_car_price.csv")
print(df.shape)
print(df.dtypes)
df.describe()
```

The dataset was confirmed to contain 1000 rows and 5 numeric columns.

4. Data Quality Issues Identified

4.1 Missing Values

Missing values were found in four columns, each affecting under 1.1% of the data:

- car_age_years — 10 missing (0.98%)
- km_driven — 11 missing (1.08%)

- engine_cc — 10 missing (0.98%)
- owners — 10 missing (0.98%)
- price_lakh — 0 missing

4.2 Duplicate Rows

20 fully duplicated rows were identified using `df.duplicated()` and confirmed by inspection.

4.3 Invalid Values

Using boolean filtering, the following invalid entries were identified:

- 5 rows with negative car_age_years (e.g., -2.0 years — impossible)
- 5 rows with negative km_driven (-10000 km — impossible)
- 5 rows with unrealistic owner counts (8 owners — treated as a data-entry error)
- 20 rows with negative or zero price_lakh — invalid for a resale price

5. Visual Analysis

5.1 Distributions and Outliers

Histograms (with KDE curves) and boxplots were used to visualize the distribution of car_age_years and km_driven. The boxplots clearly flagged the invalid negative-age and negative-km values as isolated outlier points far outside the normal range.

5.2 Feature vs. Target Relationships

Scatterplots were used to compare each key feature against price_lakh:

- Car age vs. price — a clear downward trend; older cars are worth less.
- Kilometers driven vs. price — a downward trend; higher mileage lowers price.
- Engine size vs. price — a weak, noisy relationship with no strong trend.

5.3 Correlation Heatmap

A correlation heatmap was generated using `df.corr(numeric_only=True)` and `sns.heatmap()` to numerically confirm the relationships observed visually:

Feature	Correlation with price_lakh
car_age_years	-0.67
km_driven	-0.54
engine_cc	0.13
owners	-0.01

car_age_years and km_driven showed the strongest (negative) correlations with price, confirming them as the two most important predictive features. engine_cc and owners had little to no linear relationship with price.

6. Data Cleaning

Based on the issues identified above, the dataset was cleaned in the following order:

- Removed 20 duplicate rows
- Removed rows with negative car_age_years

- Removed rows with negative km_driven
- Removed rows with unrealistic owners (> 5)
- Removed rows with price_lakh ≤ 0
- Filled remaining missing values in numeric columns with the column median (chosen over mean since it is not skewed by outliers)

Dataset size after each cleaning step:

Cleaning Step	Rows Remaining
Original dataset	1000
After removing duplicates	1000 (20 duplicates found)
After removing negative car_age_years	985
After removing negative km_driven	972
After removing unrealistic owners (>5)	957
After removing invalid price_lakh (<=0)	938

The final cleaned dataset contains 938 rows with zero missing values and zero duplicates.

7. Feature Engineering

7.1 km_per_year (required feature)

Calculated as km_driven divided by (car_age_years + 1), giving the average distance driven per year. Adding 1 to the denominator avoids division-by-zero errors for brand-new (0-year-old) cars.

```
df['km_per_year'] = df['km_driven'] / (df['car_age_years'] + 1)
```

7.2 is_high_mileage (additional feature)

A binary flag marking whether a car's km_driven is above the dataset median, helping capture threshold-based buyer behavior that a raw numeric value might not.

```
median_km = df['km_driven'].median()
df['is_high_mileage'] = (df['km_driven'] > median_km).astype(int)
```

8. Final Dataset & Train-Ready Split

After a final re-check confirmed zero missing values, zero duplicates, and realistic min/max ranges, the dataset was split into features and target:

```
X = df.drop('price_lakh', axis=1)
y = df['price_lakh']
```

X contains car_age_years, km_driven, engine_cc, owners, km_per_year, and is_high_mileage. y contains price_lakh, the target variable.

9. Conclusion

This project involved performing exploratory data analysis (EDA) and feature engineering on a used car price dataset containing 1000 records. Data quality issues found included 20 duplicate rows, missing values in four columns (~1% each), 5 rows with invalid negative car ages, 5 rows with invalid negative kilometers driven, 5 rows with unrealistic owner counts, and 20 rows with negative or zero prices. After cleaning, the dataset was reduced from 1000 to 938 valid records.

car_age_years showed the strongest relationship with price (correlation: -0.67), followed by km_driven (-0.54). engine_cc had a weak positive correlation (0.13), and owners showed almost no correlation (-0.01). Two new features, km_per_year and is_high_mileage, were engineered to make the data more meaningful for future modeling.

Overall, car age and mileage were found to be the dominant drivers of resale price in this dataset, while engine size and ownership history played a minor role.

10. What I Learned

Through this project, I learned how to perform a complete EDA and feature engineering pipeline, from loading raw data to preparing it for modeling. I learned how to inspect a dataset's structure using pandas, and how to identify issues like missing values, duplicates, and invalid entries using boolean filtering. Visualizations like histograms, boxplots, and scatter plots helped me spot distributions and outliers, while a correlation heatmap numerically confirmed that car age and kilometers driven were the strongest predictors of price. I also learned the importance of cleaning data before drawing conclusions, since invalid values can distort results, and I practiced engineering new features and separating the data into X and y for future modeling.

11. Use of AI Assistance

I used AI in a limited way, mainly to clarify a few concepts and confirm my approach at certain steps — such as understanding why certain values were invalid or why median was a better choice for filling missing data. All the code was written, run, and verified by me in Google Colab using my own understanding of the dataset and requirements.